DOF-Core vs Standard AI: 100 Experimental Rounds in Civilization

Detailed statistical report • 11 conditions × 100 seeds × 3,000 turns

1,100	complete simulations
11,000	initial-civilization observations
100	paired d0 vs d10 comparisons

Main finding: Under the 100% DOF condition, all ten initial civilizations survived in every one of the 100 games; their combined game score was 3.88 times that of the standard AI condition, with zero recorded war declarations.

Important counterfinding: When just one initial civilization adopted DOF, its survival rate dropped to 18%. With up to five DOF civilizations, average total game score was lower than with no DOF.

Source: the supplied rounds.tgz archive (100 JSON files). All statistics were computed from the supplied data. The game-engine source code and saved-game files are not included in the archive.

Report date: November 3, 2026 • English edition

1. Executive summary and principal effects

The experiment shows a **strongly nonlinear relationship** between the share of civilizations following the DOF policy and the outcomes of the simulations. Full adoption is associated with much better survival, scientific and economic indicators, and fewer wars. Partial adoption, however, may worsen the outcomes of individual DOF agents and reduce aggregate performance.

Outcome (100 paired seeds)	0 DOF	10 DOF	Change
Surviving civilizations	5.69	10.00	+4.31 of 10
Game score	14,466.1	56,122.3	+288.0%
War declarations	146.7	0.0	-100.0%
Units destroyed	5,062.4	22.1	-99.6%
Cities	111.8	129.8	+16.1%
Population	53,630.6	115,062.2	+114.5%
Techs metric	5,066.4	24,144.6	+376.6%
Arriving spaceships	1.74	8.32	+378.2%

In all 100 matched comparisons, 100% DOF exceeded the standard AI condition on score, population, techs, and arriving spaceships. It preserved more initial civilizations in 99 comparisons, with one tie. War declarations and units destroyed were lower in all 100 comparisons.

Lower conflict counts are descriptive differences, not independent proof of better strategy in every respect. The techs variable is reported as a game metric, not interpreted as the number of distinct technologies discovered.

2. Experimental design and unit of analysis

Each round contains eleven conditions d0–d10. In d0, all ten initial civilizations use the standard AI; in d10, all ten follow the DOF policy. Intermediate conditions contain exactly k DOF civilizations. Membership is verified using the **dof_roster** field, rather than inferred from the condition name.

The archive covers 100 rounds with different seeds. Within any given round, the eleven conditions share the same seed, so **seed-matched differences** are used for comparison. DOF-controlled slots rotate among rounds: every initial slot 0–9 occupies the d1 DOF position in exactly ten rounds. Moving from d(k−1) to dk changes exactly one additional slot.

Game parameters: turns=3000, aifill=10, size=2, pool=4; engine_md5=76fde78d945395ccd12c8c2a6e42516a, constant across all runs. All 1,100 simulations report final_turn=3001. Standard victory conditions were disabled: endspaceship=disabled, endvictory=disabled.

Two populations: (a) the **initial cohort**, the same ten player IDs 0–9 in every simulation; and (b) **all participants**, including additional IDs that may emerge during play. Main economic outcomes are calculated **for the initial cohort alone**, avoiding a changing denominator as extra participants appear. Wars and combat losses are whole-game aggregate metrics.

Independent unit for uncertainty estimates: the round/seed (n=100), rather than the 1,000 civilizations in each condition. Confidence intervals use nonparametric bootstrap sampling of the 100 rounds (20,000 resamples). Paired comparisons with baseline use the Wilcoxon signed-rank test, with a Holm correction across the ten comparisons for each metric. Main d0–d10 contrasts additionally use exact sign tests.

Conditions sharing a seed are not independent. Because civilizations interact, individual DOF assignments cannot be treated as isolated “treatments”: no-interference assumptions (SUTVA) are violated.

3. Survival: the central system-level effect

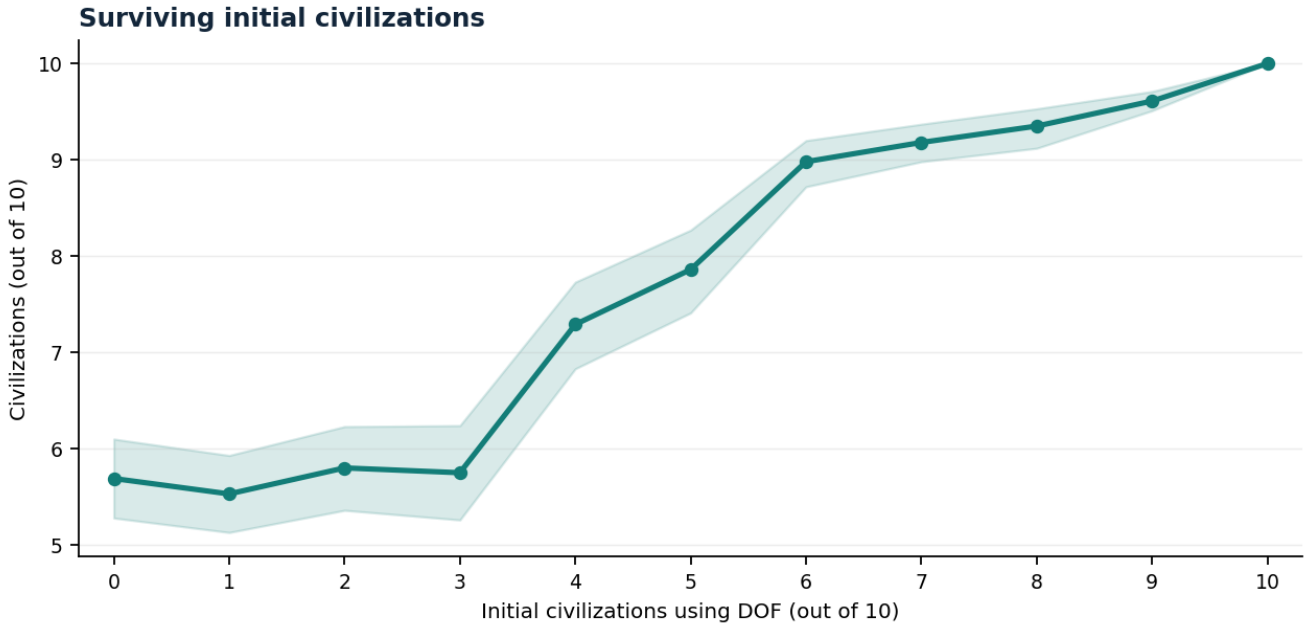


Figure 1. Average number of surviving initial civilizations at the end of 3,000 turns. Shaded bands: 95% bootstrap confidence intervals across seeds.

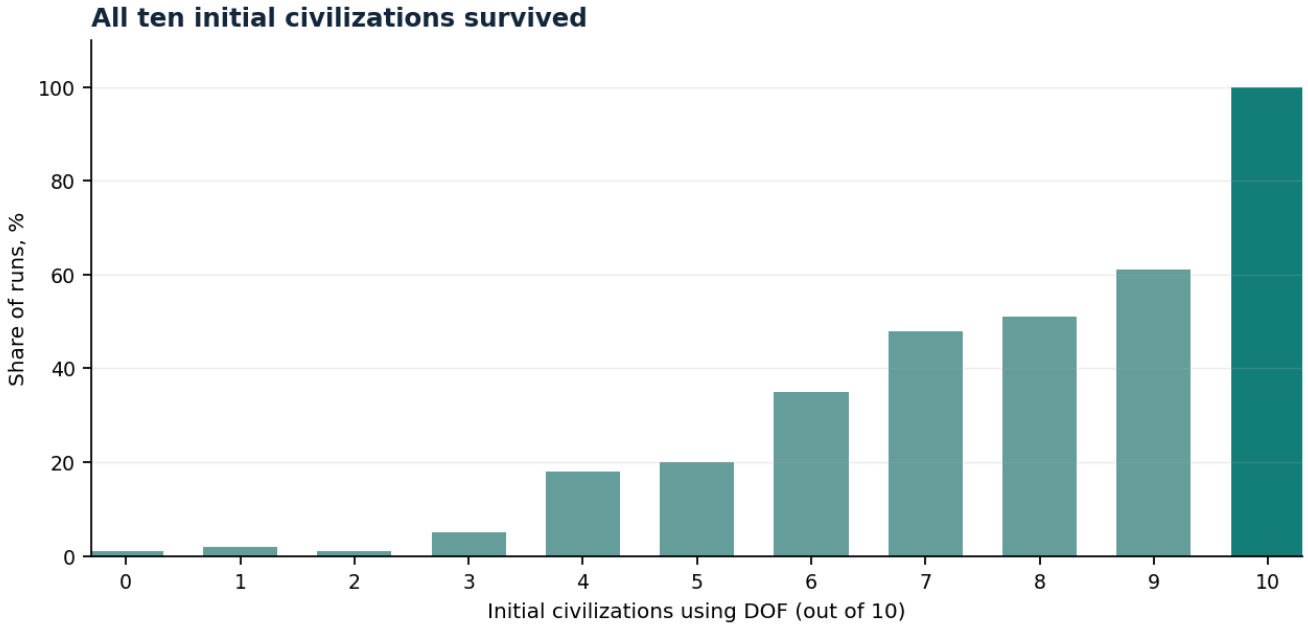


Figure 2. Share of rounds where all ten initial civilizations survived. d0: 1%; d10: 100%.

Survival improves markedly from around four to six DOF civilizations: mean survivors rise from 5.69 (d0) to 7.29 (d4), 7.86 (d5), 8.98 (d6), and ultimately 10.00 (d10). This is a description of the observed curve; a universal “threshold” is not established by these data.

4. A DOF civilization among standard AI opponents

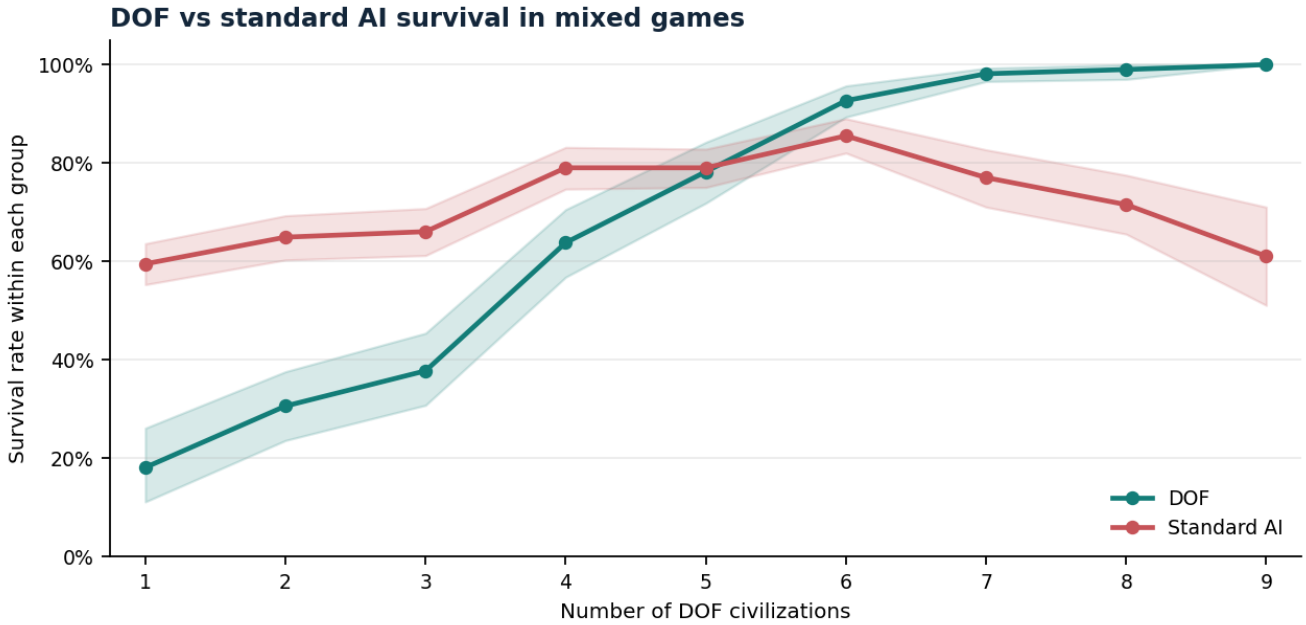


Figure 3. Within-group survival rates for DOF and standard AI, conditional on the number of DOF-controlled slots. Intervals use 100 round-level group averages.

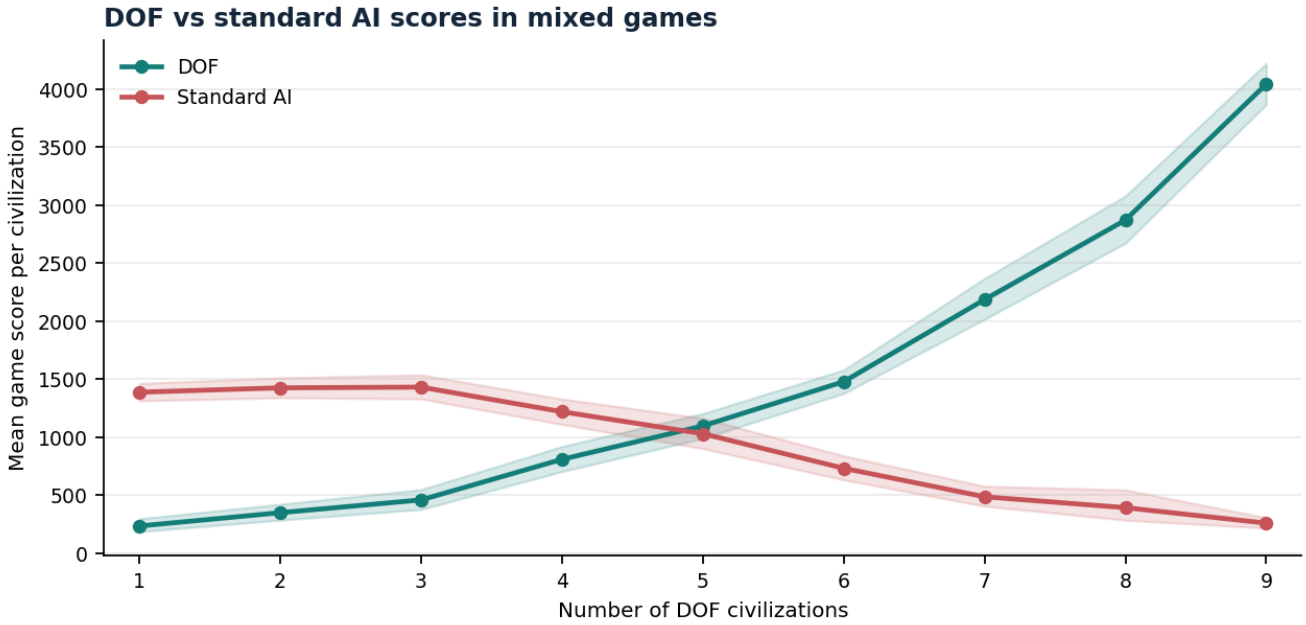


Figure 4. Mean individual final game score in mixed-policy settings (the ten original civilizations).

Under d1, the single DOF civilization survives only **18%** of games, while the nine standard AI civilizations collectively exhibit a **59.4%** survival rate. The pattern changes as DOF adoption rises: at d6, survival is **92.7%** for DOF civilizations and **85.5%** for standard AI. At d9, the DOF group survives completely, whereas the last standard AI civilization survives in only **61%** of games.

5. Game score: a U-shaped relationship

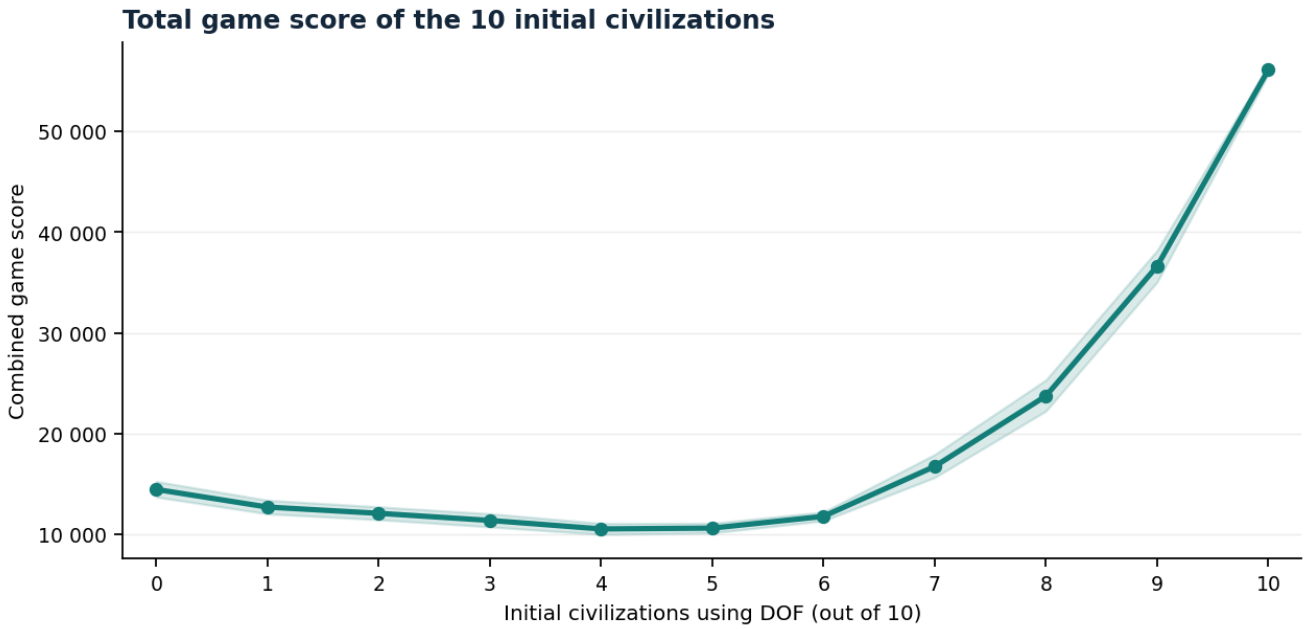


Figure 5. Average combined game score of the initial ten civilizations. d10: 56,122 versus 14,466 in d0 (3.88× higher).

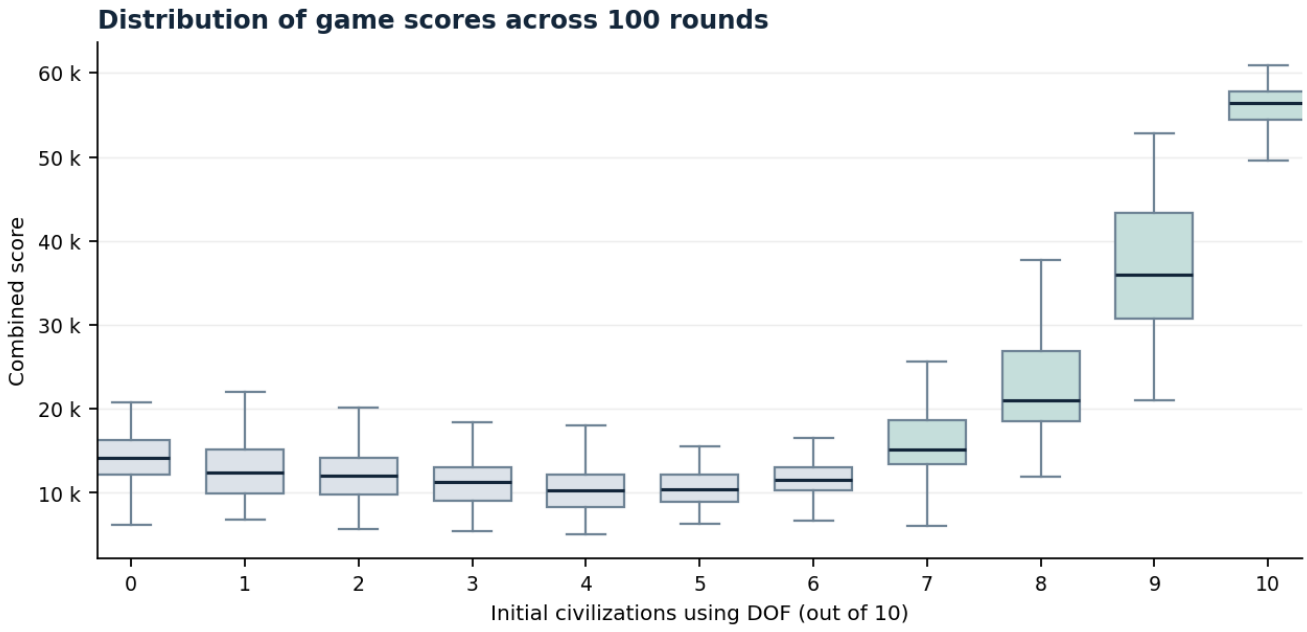


Figure 6. Distribution of final scores across 100 rounds: boxes show interquartile ranges and medians; outliers are hidden for readability.

DOF adoption does not automatically increase score. Conditions d1–d6 have lower average scores than d0, with the trough around d4–d5. d7 is higher than d0 on average but wins only **55 of the 100** paired comparisons. For d8 the figure is **92 of 100**; for both d9 and d10 it is **100 of 100**.

6. Conflict: initial escalation, then collapse

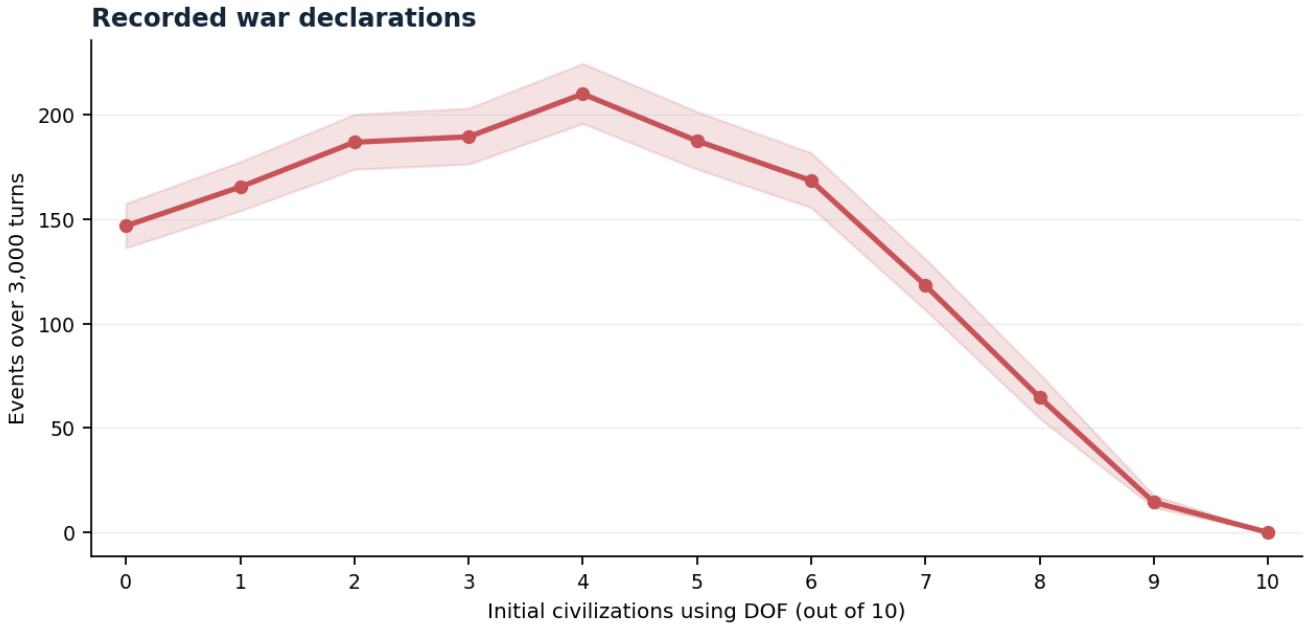


Figure 7. War declarations measured by the game variable dof_declares. Mean declarations peak at roughly four DOF civilizations; d10 has zero.

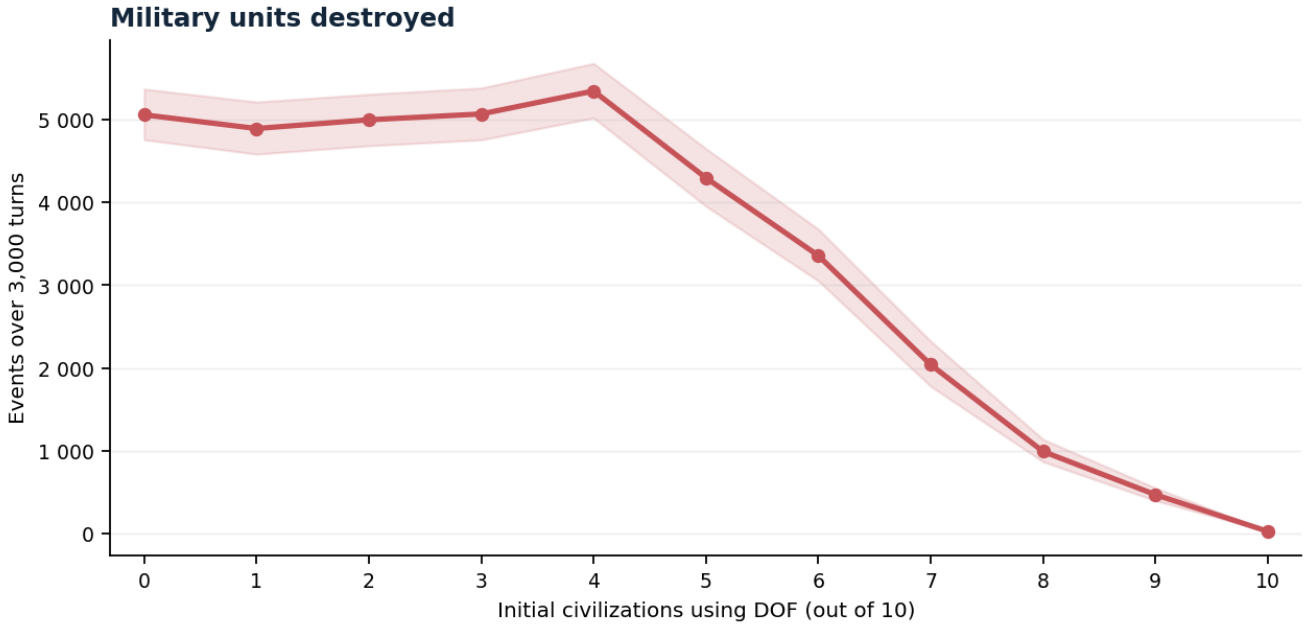


Figure 8. Units destroyed (units_killed, a whole-game measure). This is not necessarily equivalent to people killed or civilizations eliminated.

At intermediate DOF shares, conflict **can increase**: mean war declarations reach 209.9 at d4, compared with 146.7 at d0. The curve then falls sharply after d7: 118.2 → 64.6 → 14.5 → 0. This nonlinearity is consistent with a hypothesis of competing behavioral norms in mixed populations, but it does not establish the causal mechanism.

7. Science and population

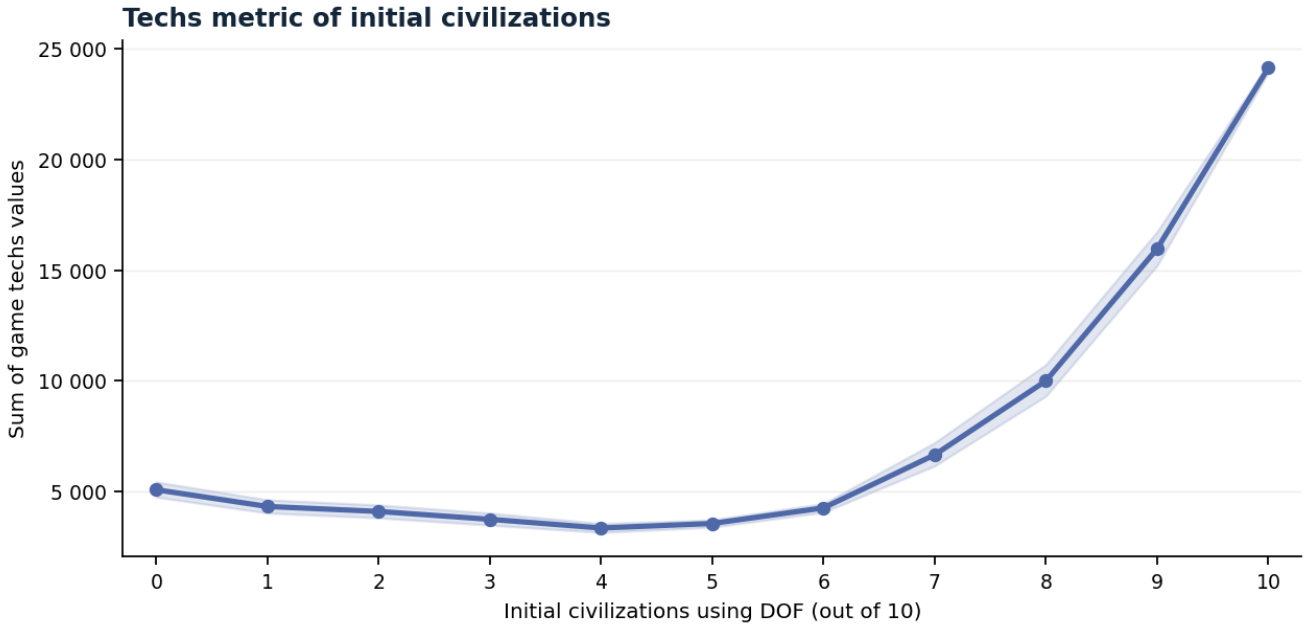


Figure 9. Sum of the techs game variable across the original ten civilizations; d10 averages 4.77× d0. It is not necessarily a count of unique discoveries.

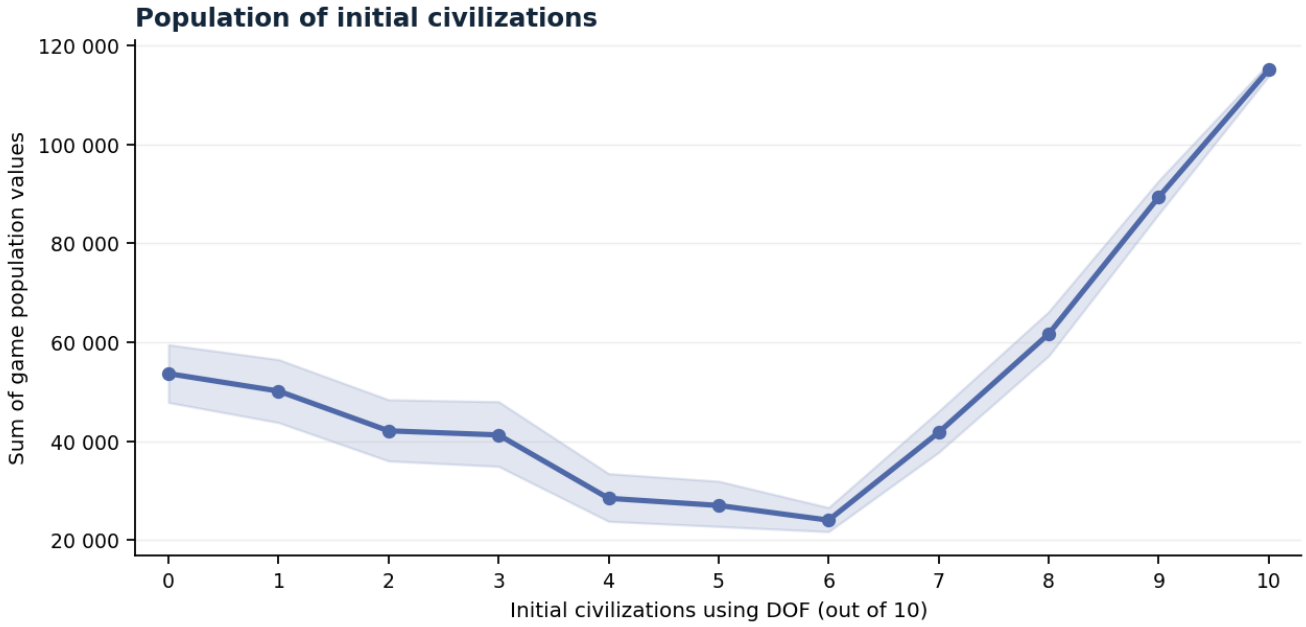


Figure 10. Total population of the original civilizations; d10 averages 2.15× d0.

Scientific and population outcomes are particularly sensitive to the degree of DOF adoption. Conditions d4–d6 frequently underperform the baseline, while clear increases occur only at high DOF saturation. These two indicators describe the **state at the final turn**, not cumulative welfare over the entire game.

8. Cities, additional players, and space milestones

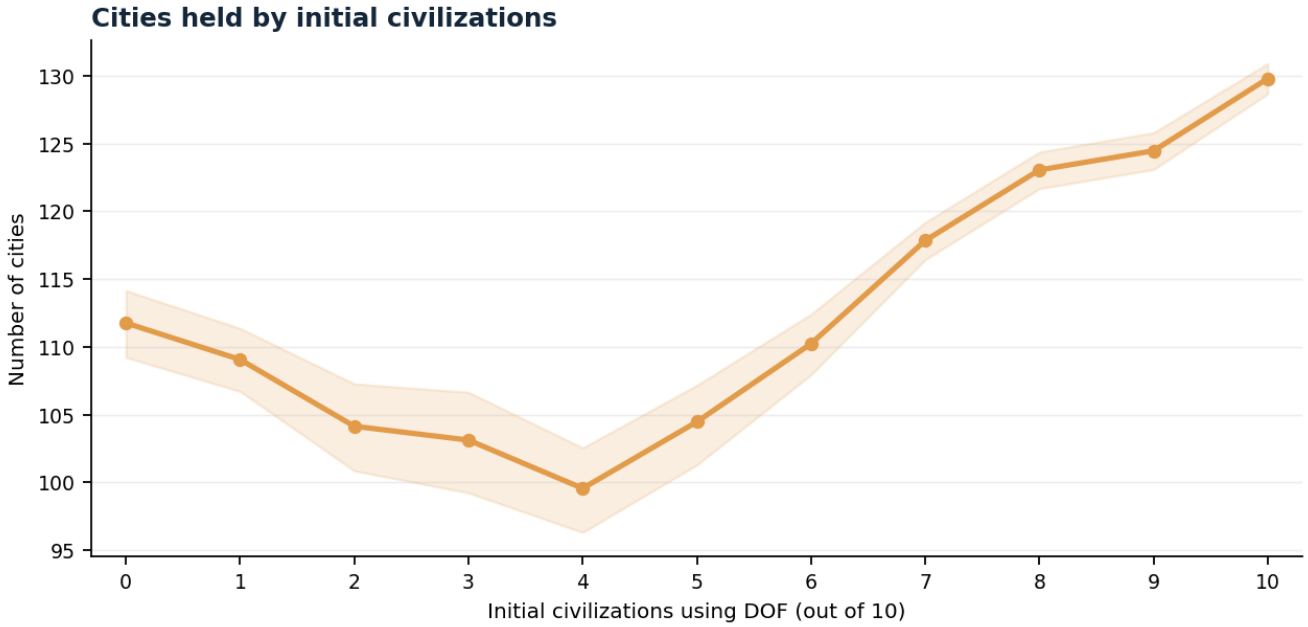


Figure 11. Number of cities held by the original ten civilizations at the final snapshot.

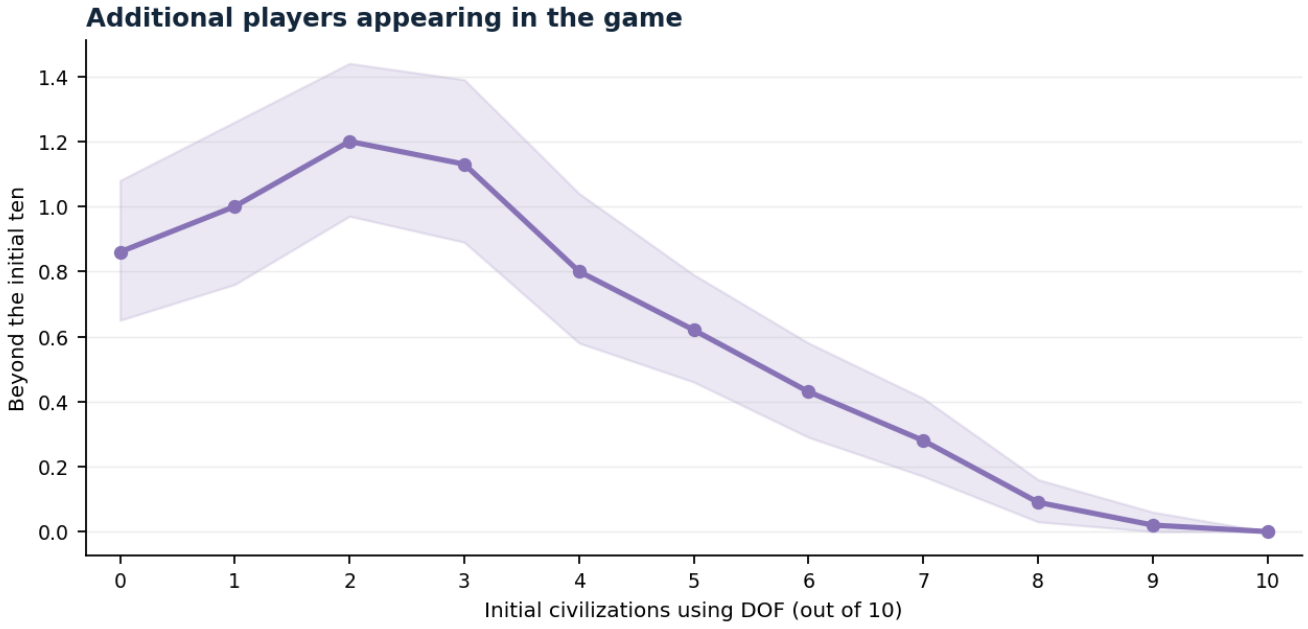


Figure 12. Mean number of additional participants (IDs outside 0–9). d0 averages ≈0.90; no additional participants were observed in d10.

Additional participants appeared in 50 of the 100 d0 rounds but none of the d10 rounds. The archive does not explain why these IDs arise. They might be linked to game events, but their specific nature should not be asserted without source-code verification.

Arriving spaceships average **1.74** in d0 and **8.32** in d10. However, space victory was disabled in the game settings, so this is a technological milestone indicator, **not a victory count**.

9. Paired differences and uncertainty

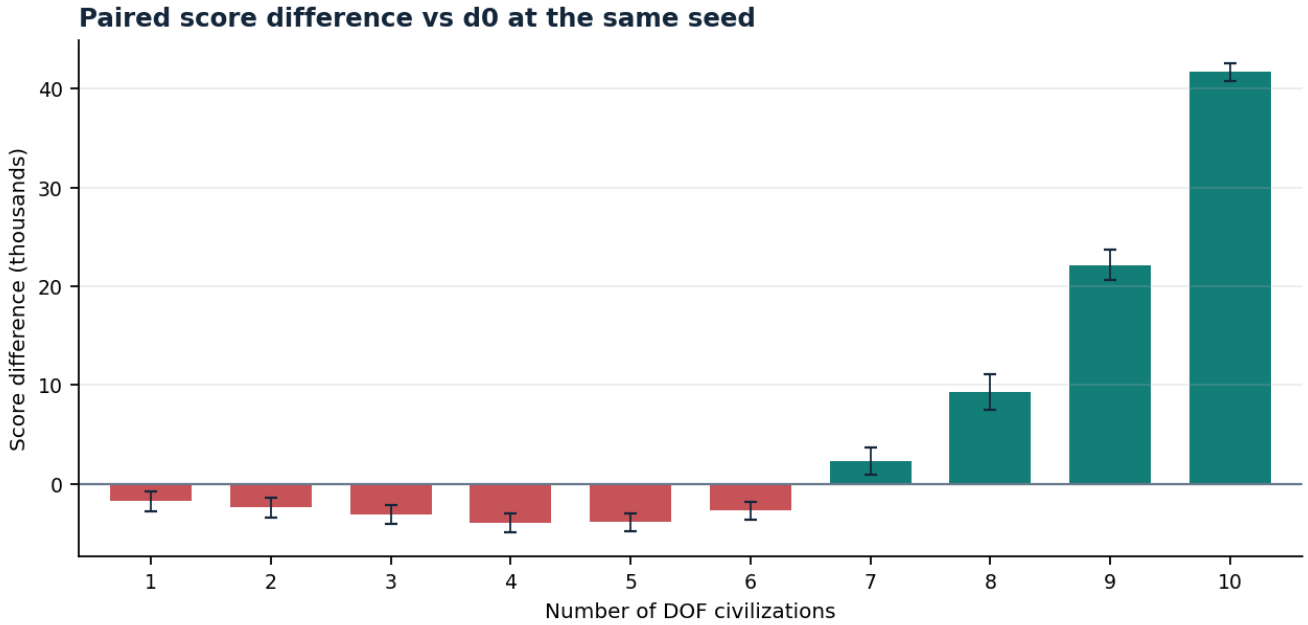


Figure 13. For each k, the average paired difference in combined score relative to d0 at the identical seed. Error bars: 95% bootstrap confidence intervals.

Outcome	Difference d10 − d0	95% confidence interval
Surviving civilizations	+4.31	[3.90, 4.72]
Game score	+41,656.2	[40,765.3, 42,517.9]
War declarations	-146.7	[-157.5, -136.3]
Units destroyed	-5,040.2	[-5,345.3, -4,730.5]
Cities	+18.0	[15.4, 20.8]
Population	+61,431.6	[55,475.1, 67,422.5]
Techs metric	+19,078.2	[18,687.6, 19,458.5]
Arriving spaceships	+6.58	[6.26, 6.89]

For all eight primary d10–d0 outcomes, the two-sided exact sign test yields Holm-adjusted $p < 10^{-20}$. For the ten intermediate-vs-baseline comparisons, Wilcoxon signed-rank tests receive separate Holm adjustments for each outcome (next section).

10. Statistics across all eleven conditions

DOF / 10	Survivors out of 10	All ten survive	Combined score	War declarations	Units destroyed
0	5.69	1%	14,466	146.7	5,062
1	5.53	2%	12,715	165.3	4,897
2	5.80	1%	12,096	186.7	5,003
3	5.75	5%	11,392	189.3	5,073
4	7.29	18%	10,546	209.9	5,351
5	7.86	20%	10,628	187.4	4,300
6	8.98	35%	11,793	168.3	3,359
7	9.18	48%	16,749	118.2	2,039
8	9.35	51%	23,743	64.6	989
9	9.61	61%	36,625	14.5	466
10	10.00	100%	56,122	0.0	22

Below, conditions d1–d10 are compared with d0 using matched seeds. Holm-adjusted p-values are calculated **separately** for score and survival. Each family comprises ten planned tests, limiting false positives from multiple comparisons.

DOF / 10	Δ score	Holm p score	Δ survivors	Holm p survival
1	-1,751	0.004	-0.16	1
2	-2,370	1.2e-04	+0.11	1
3	-3,074	4.1e-08	+0.06	1
4	-3,920	6.7e-10	+1.60	1.0e-04
5	-3,838	1.3e-10	+2.17	6.1e-08
6	-2,673	4.5e-07	+3.29	9.3e-15
7	+2,283	0.009	+3.49	2.3e-15
8	+9,277	1.0e-14	+3.66	2.9e-16
9	+22,159	3.9e-17	+3.92	9.1e-17
10	+41,656	3.9e-17	+4.31	4.6e-17

These p-values apply to the specific map generator, game algorithms and settings used in the experiment. Statistical significance does not imply that the findings generalize to human societies or real-world institutions.

11. Switching one additional civilization to DOF

Moving from d(k−1) to dk changes exactly one additional civilization slot in each round. We can therefore compare the final outcomes of the same initial civilization in two scenarios sharing the same seed. This is useful for testing how the apparent advantage of an individual DOF agent depends on the environment around it.

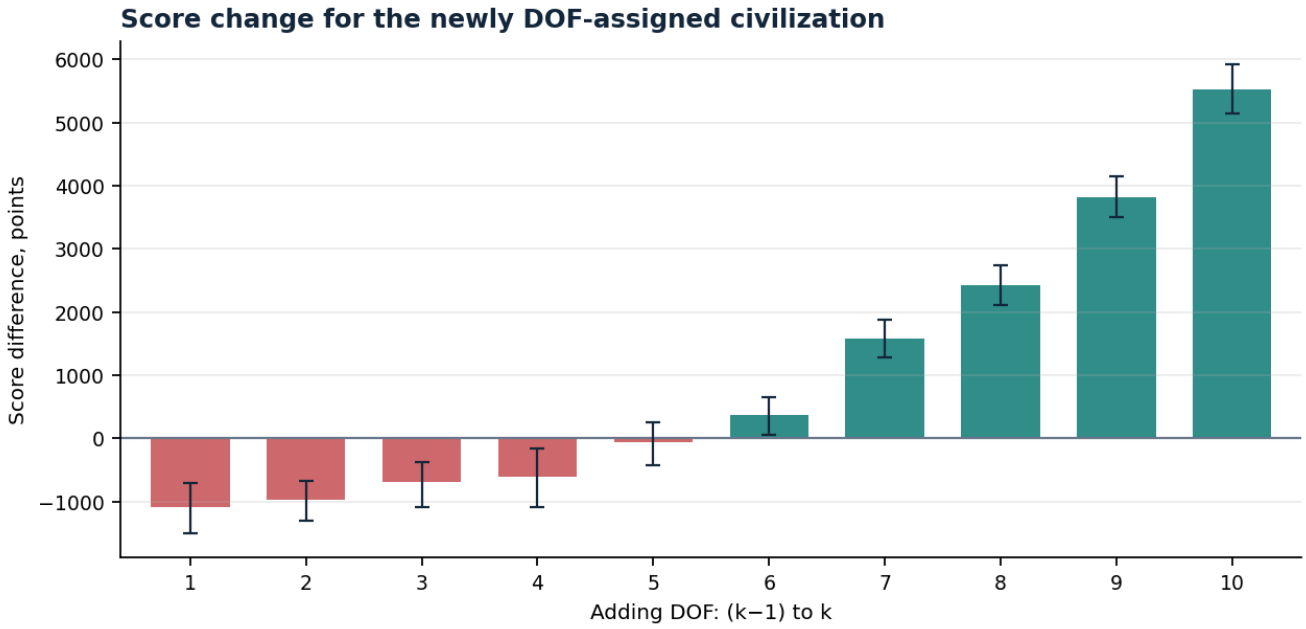


Figure 14. Score change for the civilization newly assigned DOF between d(k−1) and dk; 100 seed-matched pairs per transition.

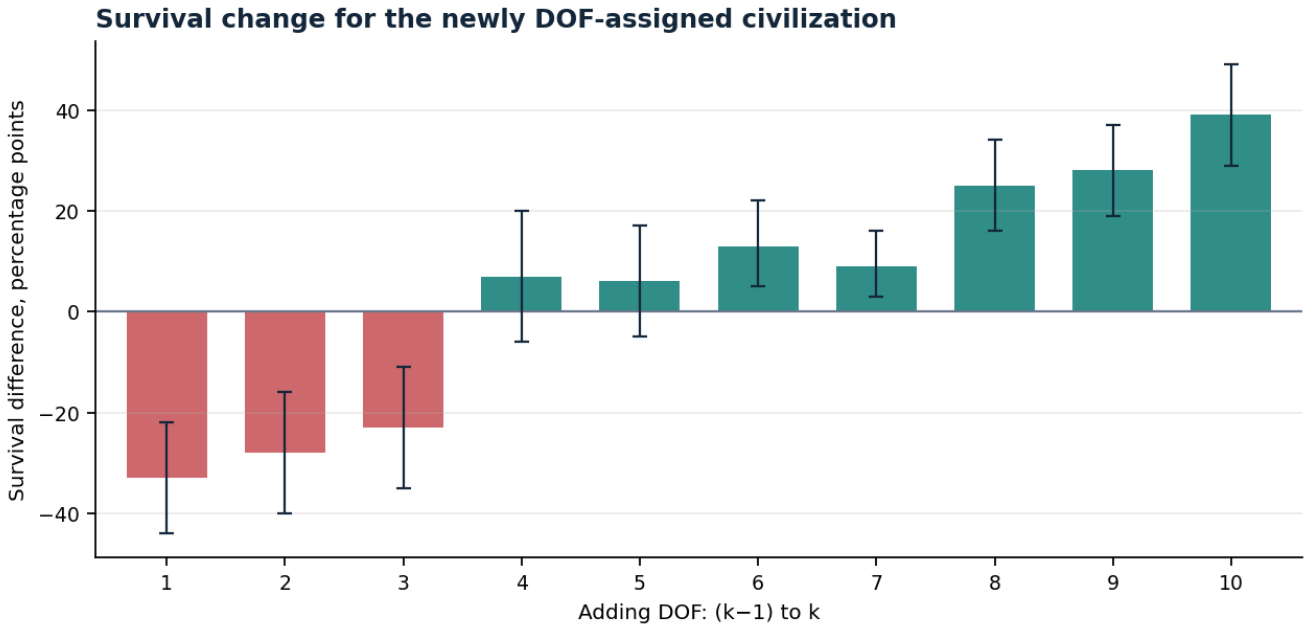


Figure 15. Change in survival probability for that same civilization (percentage points), again matched by seed.

Switching the first slot to DOF reduces its score by an average of **1,080 points** and its survival probability from **51% to 18%**, compared with the same slot in d0. The mean score effect first becomes positive when moving from d5 to d6 (+369 points), then rises sharply after d6 to d7. This is strong evidence of **context dependence**, but the estimate captures a switch **within an entire evolving simulation**, not an isolated, interaction-free benefit of the policy.

12. When civilizations disappear

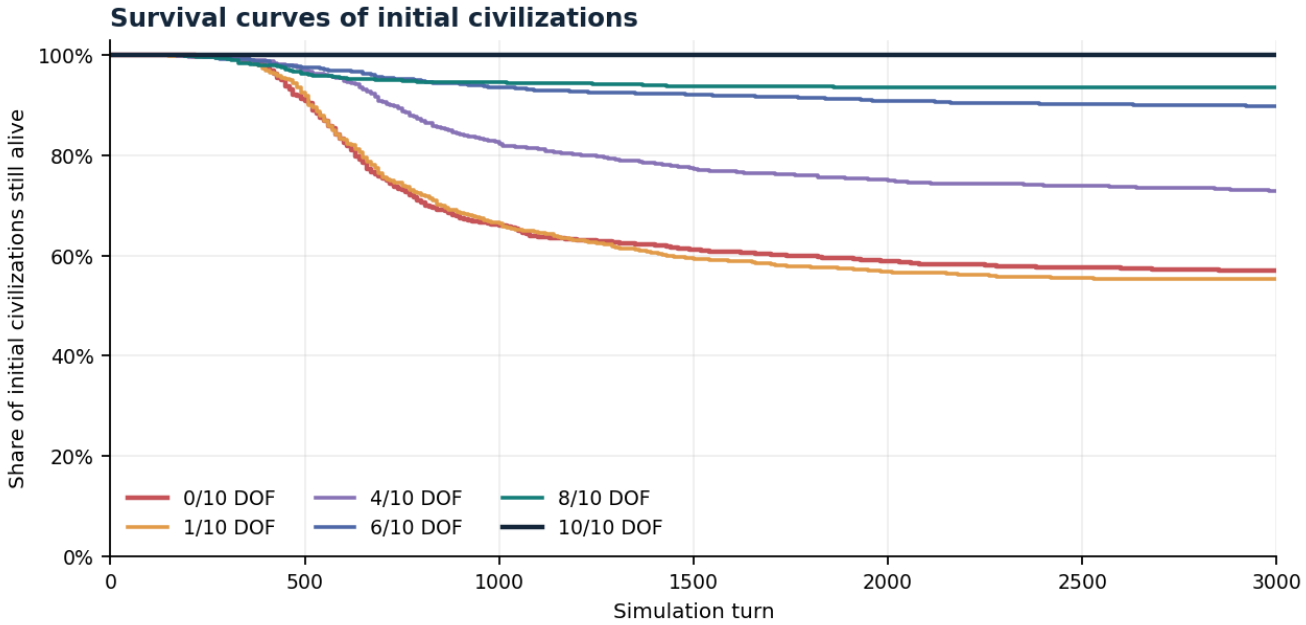


Figure 16. Share of original civilizations surviving through each turn, derived from turns_alive. Each curve pools 100 games per condition (1,000 original civilizations).

In mixed-policy conditions, some original states disappear within the first few hundred turns. Early elimination can sharply reduce the final score even if the survivors later prosper. Under d10, the survival curve remains at 100% through the end of the game: no original civilization disappeared in any of the 100 rounds.

These curves are descriptive: 1,000 civilizations in one condition do not represent 1,000 independent experiments. Confidence intervals and significance tests are based on the 100 rounds, not individual nation observations.

13. Conflict and development are correlated

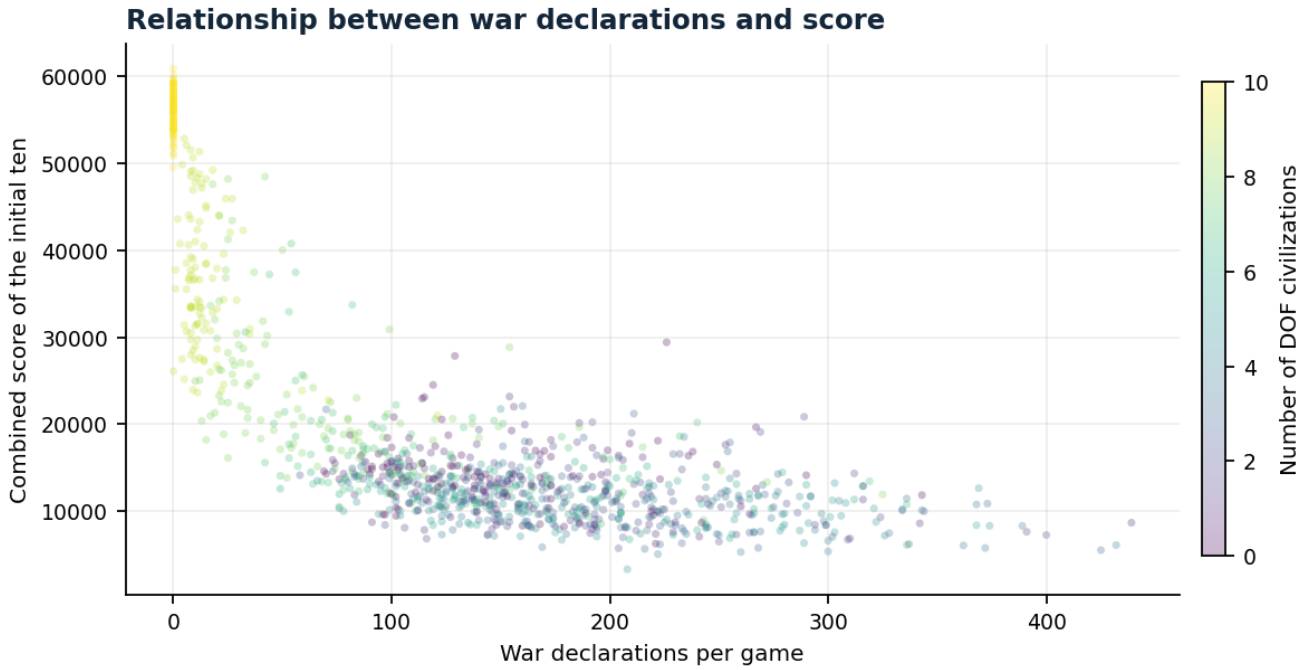


Figure 17. Each point represents one of 1,100 simulations; color indicates the number of DOF-controlled civilizations. The negative association may be driven by the common factor k.

Conflict and lower game scores are associated in this experiment, but the scatterplot does not identify causal direction. Wars may hamper development; weak development may lead to wars; or DOF rules may influence both independently. A causal claim that “fewer wars cause higher scores” cannot be made from this plot alone.

Importantly, the logs show that the DOF policy includes **active action restrictions**. The dof_blocked and dof_suppressed fields are nonzero in many scenarios (in d10, averages of approximately 5,947 and 90.6 per game, respectively). Without the source code, the meaning of these counts is uncertain. Numerous blocked attempts may reflect repeated checks of the same plan rather than the same number of distinct wars prevented.

14. Data integrity checks

Check	Result
Files / rounds	100 JSON files, 0000–0099 with no missing rounds
Conditions per round	11; 1,100 simulations total
Original players	Ten (IDs 0–9) in every condition
Player records	11,643, including 643 additional player records
DOF assignment	Exactly k original DOF slots in condition dk
Seeds within each round	One shared seed across all eleven conditions
Aggregate consistency	score_sum and alive agree with totals from nations
Simulation duration	3,000 configured turns / final_turn=3001
Victory settings	Space and standard victory disabled
Event log	declares stores at most 40 records per simulation
Saves / code	.sav.zst paths referenced but not included; no source code

Some aggregate statistics cannot be independently checked against their individual events. The declares event log is truncated after 40 records even though aggregate declaration counts reach up to 439 in some simulations. Likewise, the archive lacks complete per-turn time series for score, population, cities and science; therefore no such time-series charts were invented.

The started_utc and finished_utc timestamps are identical within each round file, so accurate wall-clock durations cannot be reconstructed from them; a separate secs field is available. This affects runtime performance analysis, not the evaluation of in-game outcomes.

15. What the data establish—and what they do not

Directly supported observations

1) Under the fixed settings used here, universal DOF adoption is exceptionally stable relative to standard AI and yields higher scores, populations, techs and initial-civilization survival. 2) DOF effects are highly **context-dependent**: one to three DOF slots do not improve aggregate survival, while their individual occupants tend to fare worse than neighboring standard AI. 3) At high levels of DOF adoption, conflict falls as game achievements rise.

Plausible but unverified explanations

Hypothesis A: a small number of DOF civilizations avoid strategies standard AI uses to gain unilateral advantages, making the former vulnerable in a competitive environment. **Hypothesis B:** once DOF becomes common enough, war costs fall and resources become available for development. **Hypothesis C:** direct action restrictions themselves generate peace under d10; without ablation experiments, the effects of banning war cannot be separated from those of the broader DOF-Core framework.

Principal limitations

The policy implementation and its compliance with DOF-SPEC cannot be verified independently because the relevant source code was not part of the supplied archive. A shared initial seed does not guarantee identical subsequent random-number paths. DOF slot assignments are **balanced, but are not an independent complete randomization of all possible compositions**. Finally, game score and techs are not direct measurements of future attainable options (DoF) as defined in the formal framework.

16. Recommendations for the next experiments

1. Policy ablations. Include “war-ban only,” “peace-first only,” “DOF without war block,” and “full DOF” conditions to distinguish direct restrictions on war from the possible benefits of the broader DOF concept.

2. Better factorial coverage. For each k, evaluate different sets of assigned slots: ideally all C(10,k) combinations, or a uniformly sampled subset. Preserve the same seeds for matched comparisons. The current design rotates slots fairly but uses only a limited family of nested compositions.

3. Turn-level measurements. Every 25–50 turns, record score, population, cities, science output, treasury, diplomatic relationships and war status for each civilization. This will make it possible to study when results diverge and whether conflict changes precede development changes.

4. Direct measures of degrees of freedom. Track available peaceful actions, autonomous agents, alternative development trajectories, decision reversibility and irreversible losses. This would test whether the policy achieves the specific goals of DOF-Core, not just conventional game outcomes.

5. Guard against pseudoreplication. Expand the number of independent rounds and vary maps and rules. Cluster uncertainty estimates by seed and slot where appropriate; publish fixed engine and patch versions, configurations, and relevant save files.

6. Validate counters. Check the precise semantics of dof_blocked, dof_suppressed, techs, bulbs_last_turn, and additional player IDs. In particular, assess whether rejected actions are repeatedly attempted and how this affects AI efficiency.

Overall conclusion: In this simulation, a fully DOF-oriented environment is substantially more successful on the reported game metrics, but this does not show that a DOF agent has an unconditional advantage in every environment. The most scientifically interesting finding is **that the apparent effect changes sign depending on how widely the policy is adopted**. This warrants further controlled research rather than a declaration that the broader theory has been proven.

Appendix A. Within-group metrics in mixed conditions

DOF / 10	DOF survival	Standard AI survival	DOF score per player	Standard AI score per player	All ten survive
0	—	56.9%	—	1,446.6	1%
1	18.0%	59.4%	234.1	1,386.8	2%
2	30.5%	64.9%	349.0	1,424.8	1%
3	37.7%	66.0%	458.9	1,430.8	5%
4	63.7%	79.0%	808.5	1,218.7	18%
5	78.2%	79.0%	1,096.3	1,029.3	20%
6	92.7%	85.5%	1,477.7	731.6	35%
7	98.1%	77.0%	2,184.4	486.1	48%
8	99.0%	71.5%	2,869.8	392.3	51%
9	100.0%	61.0%	4,040.7	259.0	61%
10	100.0%	—	5,612.2	—	100%

Interpretation: Comparing DOF with standard AI within a mixed condition is not equivalent to a causal, isolated comparison of one agent under two policies. Slot assignment is balanced, but all players affect one another, and changes in one policy alter the behavior of others.

Appendix B. Variables, exclusions, and reproducibility

Key fields: sections.d0…d10, policy, seed, dof_roster, nations[id, is_alive, turns_alive, score, cities, population, techs, bulbs_last_turn], dof_declares, units_killed, totals, spaceship_arrived, victory_settings. Main score = Σ nations.score for IDs 0–9 only. Survival = Σ is_alive for IDs 0–9 only.

Uncertainty estimation: 20,000 seed-level bootstrap resamples with replacement (fixed RNG seed 20261103). Paired intervals were computed from differences dk–d0. Significance was assessed using a two-sided exact sign test for eight primary outcomes and paired Wilcoxon tests for ten comparisons per outcome; Holm correction was applied within each declared family.

Source: rounds.tgz, SHA-256: 5c6d8ec4f45da9d9c6574a4418a758fc7ce1a0c8c17efb9b937d377ace6dabb4

Reference repository link (source content not used in the statistical calculations): https://github.com/UncleAndy/dof-model-civ/tree/v0.11/experiments/dof-ab-11arms-100x3000/rounds